# Daily Research Progress Report – 22 August 2026

## Toward a microscopic dynamical theory of $s\bar{s} \to \Lambda\bar{\Lambda}$ spin correlations

Reconstructed from the validated project state and the 22 August derivations

23 August 2026


**Abstract**

This report reconstructs the validated research progress for 22 August 2026 after the automatically generated daily artifact failed to persist. The strongest result of the day is a microscopic selection rule for the STAR scalar spin-correlation observable: a broad class of exchange-symmetric coherent two-spin Hamiltonians cannot change $\mathrm{Tr}\, C = \langle \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \rangle$. Therefore the measured loss of $\Lambda\bar{\Lambda}$ correlation requires singlet–triplet conversion driven by *relative* spin fields, exchange-antisymmetric interactions, or genuinely dissipative/collisional terms. This result provides a sharper target for a future two-particle Wigner/Kadanoff–Baym derivation. We also review how this result interfaces with the ${}^3P_0$ source, feed-down, the STAR observable, and existing quantum-kinetic literature.


## 1 Status and scope

The long-term objective is to construct a calculational chain

$$\boxed{\rho_{s\bar{s}}^{\mathrm{source}} \longrightarrow \text{microscopic spin transport} \longrightarrow \text{hadronization / feed-down} \longrightarrow \rho_{\Lambda\bar{\Lambda}}^{\mathrm{obs}}} \tag{1}$$

that explains the measured $\Lambda\bar{\Lambda}$ spin correlation and its dependence on pair separation.

The STAR measurement in $pp$ collisions reports a short-range relative polarization

$$P_{\Lambda\bar{\Lambda}}^{\mathrm{STAR}} = 0.181 \pm 0.035_{\mathrm{stat}} \pm 0.022_{\mathrm{sys}}, \tag{2}$$

with the correlation decreasing substantially at larger angular separation. The experimental observable is naturally written in terms of the two-spin correlation tensor $C_{ij}$,

$$\rho = \frac{1}{4}\left[\mathbb{I} \otimes \mathbb{I} + P_i \sigma_i \otimes \mathbb{I} + \bar{P}_j \mathbb{I} \otimes \sigma_j + C_{ij} \sigma_i \otimes \sigma_j\right]. \tag{3}$$

For the rotationally averaged scalar used by STAR,

$$\boxed{P_{\Lambda\bar{\Lambda}} = \frac{1}{3}\,\mathrm{Tr}\, C = \frac{1}{3}\,\langle \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \rangle\,.} \tag{4}$$

Equation (4) is the central quantity whose microscopic time evolution we studied on 22 August.

## 2 Source state: what remains robust

A vacuum-like pair-creation mechanism with $J^{PC} = 0^{++}$ naturally leads to a ${}^3P_0$ state,

$$L = 1, \qquad S = 1, \qquad J = 0. \tag{5}$$

The exact coupled state is

$$|J = 0, M = 0\rangle = \frac{1}{\sqrt{3}} \sum_{m=-1}^{1} (-1)^{1-m} |L = 1, m\rangle |S = 1, -m\rangle. \tag{6}$$

If the orbital magnetic quantum number is completely unobserved, tracing over orbital space gives

$$\rho_S = \mathrm{Tr}_L\, |J=0\rangle\langle J=0| \tag{7}$$

$$= \frac{1}{3}\sum_{m=-1}^{1} |1,m\rangle\langle 1,m| \tag{8}$$

$$= \frac{1}{3}P_{S=1}. \tag{9}$$

For two spin-1/2 particles,

$$P_{S=1} = \frac{3\mathbb{I} + \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2}{4}, \tag{10}$$

so

$$\rho_S = \frac{1}{4}\left[\mathbb{I}\otimes\mathbb{I} + \frac{1}{3}\sum_i \sigma_i\otimes\sigma_i\right]. \tag{11}$$

Thus

$$C_{ij}^{\mathrm{avg}} = \frac{1}{3}\delta_{ij}, \qquad P_{s\bar{s}}^{(0)} = \frac{1}{3}. \tag{12}$$

The distinction between this orbitally averaged state and a momentum-conditioned local spin state remains important. A future event-by-event theory should keep the orbital/fragmentation axis until the experimental projection is made.

## 3 Exact coherent two-spin evolution

### 3.1 Most general bilinear Hamiltonian

We consider

$$H = H_1 + H_2 + H_{12}, \tag{13}$$

with

$$H_1 = \frac{1}{2}\Omega_a\,\sigma_a\otimes\mathbb{I}, \tag{14}$$

$$H_2 = \frac{1}{2}\bar{\Omega}_b\,\mathbb{I}\otimes\sigma_b, \tag{15}$$

$$H_{12} = \frac{1}{4}J_{ab}\,\sigma_a\otimes\sigma_b. \tag{16}$$

Here $\boldsymbol{\Omega}$ and $\bar{\boldsymbol{\Omega}}$ are effective local precession frequencies for the $s$ and $\bar{s}$, while $J_{ab}$ is the most general real bilinear spin-coupling tensor compatible with Hermiticity.

The density matrix evolves exactly according to

$$\dot{\rho} = -i[H,\rho]. \tag{17}$$

No Markov approximation or master-equation ansatz is used at this stage.

The correlation tensor is

$$C_{ij} = \mathrm{Tr}\left[\rho\,\sigma_i\otimes\sigma_j\right]. \tag{18}$$

Differentiating,

$$\dot{C}_{ij} = \mathrm{Tr}\left[\dot{\rho}\,\sigma_i\otimes\sigma_j\right] \tag{19}$$

$$= -i\,\mathrm{Tr}\left[[H,\rho]\,\sigma_i\otimes\sigma_j\right]. \tag{20}$$

Using cyclicity of the trace,

$$\mathrm{Tr}([H,\rho]O) = \mathrm{Tr}(\rho[O,H]), \tag{21}$$

we obtain

$$\dot{C}_{ij} = -i\left\langle[\sigma_i\otimes\sigma_j, H]\right\rangle. \tag{22}$$

### 3.2 Local-field contribution

Using

$$[\sigma_i, \sigma_a] = 2i\epsilon_{iak}\sigma_k, \tag{23}$$

the first-spin term gives

$$-i\left[\sigma_i \otimes \sigma_j, \frac{1}{2}\Omega_a\sigma_a \otimes \mathbb{I}\right] = -i\frac{\Omega_a}{2}[\sigma_i, \sigma_a] \otimes \sigma_j \tag{24}$$

$$= \Omega_a\epsilon_{iak}\,\sigma_k \otimes \sigma_j. \tag{25}$$

Therefore

$$\dot{C}_{ij}\Big|_1 = \epsilon_{iak}\Omega_a C_{kj}. \tag{26}$$

Similarly,

$$\dot{C}_{ij}\Big|_2 = \epsilon_{jbl}\bar{\Omega}_b C_{il}. \tag{27}$$

### 3.3 Bilinear interaction contribution

For the interaction term,

$$[\sigma_i \otimes \sigma_j, \sigma_a \otimes \sigma_b] = [\sigma_i, \sigma_a] \otimes \frac{\{\sigma_j, \sigma_b\}}{2} + \frac{\{\sigma_i, \sigma_a\}}{2} \otimes [\sigma_j, \sigma_b]. \tag{28}$$

Since

$$\{\sigma_i, \sigma_a\} = 2\delta_{ia}\mathbb{I}, \tag{29}$$

we find

$$[\sigma_i \otimes \sigma_j, \sigma_a \otimes \sigma_b] = 2i\epsilon_{iak}\delta_{jb}\,\sigma_k \otimes \mathbb{I} \tag{30}$$

$$+ 2i\delta_{ia}\epsilon_{jbl}\,\mathbb{I} \otimes \sigma_l. \tag{31}$$

Multiplying by $J_{ab}/4$ and by $-i$ gives

$$\dot{C}_{ij}\Big|_{12} = \frac{1}{2}\epsilon_{iak}J_{aj}P_k + \frac{1}{2}\epsilon_{jbl}J_{ib}\bar{P}_l. \tag{32}$$

Combining all pieces,

$$\boxed{\begin{aligned}\dot{C}_{ij} &= \epsilon_{iak}\Omega_a C_{kj} + \epsilon_{jbl}\bar{\Omega}_b C_{il} \\ &\quad + \frac{1}{2}\epsilon_{iak}J_{aj}P_k + \frac{1}{2}\epsilon_{jbl}J_{ib}\bar{P}_l.\end{aligned}} \tag{33}$$

This is an exact coherent evolution equation for the complete tensor.

## 4 Protection theorem for the STAR scalar

Define

$$Q \equiv \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 = \sum_i \sigma_i \otimes \sigma_i. \tag{34}$$

Then

$$3P = \langle Q \rangle. \tag{35}$$

Introduce the spin-exchange operator

$$\mathsf{P}_{12} = \frac{1}{2}(\mathbb{I} + Q), \tag{36}$$

so

$$Q = 2\mathsf{P}_{12} - \mathbb{I}. \tag{37}$$

Therefore

$$[H, Q] = 2[H, \mathsf{P}_{12}]. \tag{38}$$

If the Hamiltonian is invariant under spin exchange,

$$\mathsf{P}_{12} H \mathsf{P}_{12} = H, \tag{39}$$

then

$$[H, \mathsf{P}_{12}] = 0 \quad \Longrightarrow \quad [H, Q] = 0. \tag{40}$$

Hence

$$\boxed{\frac{d}{dt}\langle Q\rangle = 0} \tag{41}$$

and therefore

$$\boxed{\frac{d}{dt}\,\mathrm{Tr}\,C = 0.} \tag{42}$$

For the Hamiltonian in Eq. (16), exchange invariance requires

$$\boxed{\mathbf{\Omega} = \bar{\mathbf{\Omega}}, \qquad J_{ij} = J_{ji}.} \tag{43}$$

Thus any common coherent precession and any exchange-symmetric bilinear interaction can rotate or redistribute tensor components but cannot change the scalar measured by STAR.

This includes, in particular,

$$H_{\mathrm{ex}} = J(r)\,\boldsymbol{S}_1 \cdot \boldsymbol{S}_2, \tag{44}$$

as well as symmetric anisotropic couplings

$$H_{\mathrm{sym}} = S_1^i J_{ij}^{(S)} S_2^j, \qquad J_{ij}^{(S)} = J_{ji}^{(S)}. \tag{45}$$

## 5 Singlet probability: a sharper physical interpretation

For two spin-1/2 particles the singlet projector is

$$\Pi_S = |S\rangle\langle S| = \frac{1}{4}\left(\mathbb{I} - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2\right). \tag{46}$$

Therefore

$$p_S = \mathrm{Tr}(\rho \Pi_S) \tag{47}$$

$$= \frac{1}{4}\left[1 - \langle Q\rangle\right] \tag{48}$$

$$= \boxed{\frac{1-3P}{4}}. \tag{49}$$

This relation reframes the problem.

If the initial source is purely triplet,

$$p_S(0) = 0, \qquad P(0) = \frac{1}{3}. \tag{50}$$

A reduction in $P$ corresponds to a growth of singlet weight.

Using the STAR central value only as a hadron-level diagnostic,

$$p_S^{\text{obs}} = \frac{1 - 3(0.181)}{4} \simeq 0.114. \tag{51}$$

This number is not yet the primordial partonic singlet probability because hadronization and feed-down have not been unfolded. Its value simply illustrates the physical content of the measured scalar.

The key microscopic question therefore becomes

> Which QCD interactions can transfer probability between the triplet and singlet sectors?

## 6 Which coherent terms can mix singlet and triplet?

Decompose the local frequencies as

$$\boldsymbol{\Omega}_c = \frac{\boldsymbol{\Omega} + \bar{\boldsymbol{\Omega}}}{2}, \qquad \delta\boldsymbol{\Omega} = \frac{\boldsymbol{\Omega} - \bar{\boldsymbol{\Omega}}}{2}. \tag{52}$$

Then

$$H_{\text{local}} = \boldsymbol{\Omega}_c \cdot (\boldsymbol{S}_1 + \boldsymbol{S}_2) + \delta\boldsymbol{\Omega} \cdot (\boldsymbol{S}_1 - \boldsymbol{S}_2). \tag{53}$$

The first term is exchange symmetric; the second is exchange antisymmetric.

For a differential field along $z$,

$$H_\delta = \frac{\delta\Omega_z}{2}(\sigma_{1z} - \sigma_{2z}). \tag{54}$$

Using

$$|T_0\rangle = \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \tag{55}$$

$$|S\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}, \tag{56}$$

we obtain

$$(\sigma_{1z} - \sigma_{2z})|\uparrow\downarrow\rangle = 2|\uparrow\downarrow\rangle, \tag{57}$$

$$(\sigma_{1z} - \sigma_{2z})|\downarrow\uparrow\rangle = -2|\downarrow\uparrow\rangle. \tag{58}$$

Therefore

$$\boxed{(\sigma_{1z} - \sigma_{2z})|T_0\rangle = 2|S\rangle.} \tag{59}$$

The two-state Hamiltonian in the $\{|T_0\rangle, |S\rangle\}$ basis is proportional to

$$H_\delta \sim \delta\Omega_z \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{60}$$

Starting from $|T_0\rangle$,

$$p_S(t) = \sin^2(\delta\Omega_z t), \tag{61}$$

and since

$$P = \frac{1 - 4p_S}{3}, \tag{62}$$

we obtain

$$\boxed{P(t) = \frac{1}{3}\left[1 - 4\sin^2(\delta\Omega_z t)\right].} \tag{63}$$

Hence *relative* spin fields are capable of coherent singlet–triplet conversion.

### 6.1 Exchange-antisymmetric bilinear interaction

Decompose

$$J_{ij} = J_{ij}^{(S)} + J_{ij}^{(A)}, \tag{64}$$

with

$$J_{ij}^{(A)} = \frac{1}{2}(J_{ij} - J_{ji}). \tag{65}$$

Any antisymmetric rank-two tensor can be written as

$$J_{ij}^{(A)} = \epsilon_{ijk} D_k. \tag{66}$$

Then

$$\begin{aligned} H_A &= S_1^i J_{ij}^{(A)} S_2^j &(67)\\ &= D_k \epsilon_{ijk} S_1^i S_2^j &(68)\\ &= \boxed{\boldsymbol{D} \cdot (\boldsymbol{S}_1 \times \boldsymbol{S}_2)}. &(69) \end{aligned}$$

This Dzyaloshinskii–Moriya-like term is odd under exchange and can also mix the singlet and triplet sectors.

## 7 Implication for the kinetic theory

The coherent calculation suggests that the most economical coarse-grained variable for the scalar measurement is the singlet probability rather than a generic "decoherence parameter."

A minimal kinetic description is

$$\boxed{\dot{p}_S = \Gamma_{T\to S} p_T - \Gamma_{S\to T} p_S + S_S^{\text{prod}},} \tag{70}$$

where

$$p_T = 1 - p_S \tag{71}$$

if leakage outside the two-spin Hilbert space is neglected.

Using Eq. (49),

$$P = \frac{1 - 4p_S}{3}, \tag{72}$$

so

$$\begin{aligned} \dot{P} &= -\frac{4}{3}\dot{p}_S &(73)\\ &= -\frac{4}{3}\left[\Gamma_{T\to S}(1 - p_S) - \Gamma_{S\to T} p_S + S_S^{\text{prod}}\right]. &(74) \end{aligned}$$

This explicitly separates singlet–triplet conversion from source preparation.

The central microscopic task is now to derive

$$\Gamma_{T\to S}, \qquad \Gamma_{S\to T} \tag{75}$$

from a two-particle nonequilibrium QCD kernel.

## 8 Connection to the Wigner/Kadanoff–Baym route

Relativistic quantum kinetic theory for massive quarks starts with a gauge-covariant Wigner function, which can be decomposed in the Clifford basis as

$$W = \frac{1}{4}\left[\mathcal{F} + i\gamma^5\mathcal{P} + \gamma^\mu\mathcal{V}_\mu + \gamma^\mu\gamma^5\mathcal{A}_\mu + \frac{1}{2}\sigma^{\mu\nu}\mathcal{S}_{\mu\nu}\right]. \tag{76}$$

The axial component $\mathcal{A}^\mu$ carries spin information.

The one-body quantum kinetic literature derives color-singlet and color-octet equations for $\mathcal{V}^\mu$ and $\mathcal{A}^\mu$, including diffusion and color-field-correlation source terms. This provides the correct relativistic starting point for strange-quark spin transport.

For the present problem, however, we require the two-particle quantity

$$W^{(2)}(1,2) = \left\langle \widehat{W}_s(1) \otimes \widehat{W}_{\bar{s}}(2) \right\rangle. \tag{77}$$

The target connected axial-axial correlator is schematically

$$\mathcal{C}^{\mu\nu}(1,2) = \langle \mathcal{A}^\mu_s(1)\mathcal{A}^\nu_{\bar{s}}(2)\rangle_c\,. \tag{78}$$

Projecting this correlator onto singlet/triplet projectors should yield the microscopic gain/loss terms in Eq. (70). In a Kadanoff–Baym formulation, these rates must arise from self-energy/collision kernels that are odd under relative-spin exchange or otherwise distinguish the two spin histories.

## 9 Relation to existing literature

The STAR observation establishes a short-range positive $\Lambda\bar{\Lambda}$ correlation and a substantial reduction with angular separation. The nonrelativistic SU(6) interpretation assumes the $\Lambda$ spin is carried by the strange quark, motivating the use of hyperon spin as a probe of ancestral strange-quark spin.

Barata et al. model the correlation loss with a spatially correlated depolarizing open-system description. Their result that relative environmental noise controls the decay of the scalar is consistent with the algebraic protection theorem above: common-mode spin rotations preserve $\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$.

Yang's quantum kinetic theory derives massive-quark spin transport from the Wigner-function approach and obtains spin-diffusion and source terms determined by color-field correlators. It therefore supplies a first-principles language for replacing phenomenological spin-noise rates.

Kumar, Müller and Yang further show that glasma color fields can generate nonzero quark–antiquark spin correlations even when single-particle polarization vanishes. Such a mechanism belongs to the production/source sector and should not be conflated with later singlet–triplet decoherence.

## 10 Hadronization and feed-down remain indispensable

Even after the partonic singlet–triplet kinetics is derived, the observable tensor is

$$C^{\Lambda\bar{\Lambda}}_{ij} = \sum_{ab} R_{ab}\,(A_a)_{ik}\,C^{s\bar{s},(ab)}_{kl}\,(\bar{A}_b)_{jl}, \tag{79}$$

where $a, b$ label production/feed-down channels and $A_a, \bar{A}_b$ are channel-specific spin-transfer maps.

The key lesson from earlier cycles remains valid: feed-down cannot be represented safely by a single universal scalar dilution if different channels have different tensor transfer coefficients.

This means that the singlet fraction inferred directly from Eq. (2) is only an observable-level diagnostic. A quantitative partonic interpretation requires Eq. (79).

## 11 What was genuinely advanced on 22 August

The main advances of the day can be stated compactly.

1. The full coherent tensor equation, Eq. (33), was derived directly from the von Neumann equation.
2. A protection theorem, Eq. (42), was established: exchange-symmetric coherent spin dynamics preserves the STAR scalar.
3. The measured scalar was reinterpreted as a singlet probability, $p_S = (1 - 3P)/4$.
4. The microscopic target was narrowed from generic "decoherence" to *singlet–triplet conversion.*
5. Differential local fields and exchange-antisymmetric bilinear interactions were identified explicitly as coherent mechanisms capable of such conversion.
6. The future Wigner/Kadanoff–Baym derivation should therefore target $\Gamma_{T\to S}$ and $\Gamma_{S\to T}$ rather than an unconstrained scalar damping constant.

## 12 Open problems

1. Derive the two-particle axial–axial Kadanoff–Baym equation with explicit color factors.
2. Identify which QCD self-energy structures generate exchange-antisymmetric or relative-spin kernels.
3. Determine whether confinement/string-breaking dynamics yields coherent oscillatory $T \leftrightarrow S$ conversion, irreversible conversion, or both.
4. Construct the channel-resolved spin-transfer matrices for primary $\Lambda$, $\Sigma^0$, $\Xi$, and mixed feed-down pairs.
5. Retain the local fragmentation/orbital axis long enough to distinguish conditional ${}^3P_0$ tensor structure from the fully orbitally averaged triplet mixture.
6. Fit tensor-level observables $C_T(\Delta R)$ and $C_L(\Delta R)$ once experimentally available, rather than only their trace combination.

## 13 Highest-value next derivation

The next calculation should start from a two-particle Schwinger–Keldysh Green function

$$G^{<(2)}(1, 2; 1', 2') \tag{80}$$

and derive coupled equations acting on the $s$ and $\bar{s}$ legs.

After Wigner transformation and a controlled gradient/quasiparticle expansion, the collision kernel should be projected with

$$\Pi_S = \frac{1}{4}(\mathbb{I} - \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) \tag{81}$$

and

$$\Pi_T = \mathbb{I} - \Pi_S. \tag{82}$$

The immediate target is

$$\boxed{\Gamma_{T\to S} \sim \mathrm{Tr}[\Pi_S\, \mathcal{C}_{\mathrm{coll}}[\Pi_T]]\,, \qquad \Gamma_{S\to T} \sim \mathrm{Tr}[\Pi_T\, \mathcal{C}_{\mathrm{coll}}[\Pi_S]]\,.} \tag{83}$$

Once these rates are obtained from QCD field correlators or a specified microscopic model, the measured $\Delta R$ dependence can be interpreted as an actual singlet–triplet conversion history rather than a generic phenomenological loss factor.

## References


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4. J. Barata, I. Cuntín, E. Rico and B. Wu, "$\Lambda\bar{\Lambda}$ spin correlations in high-energy collisions from quantum channels: an open quantum system view of hadronization," arXiv:2606.30737 (2026).


## Artifact note

The scheduled daily synthesis did execute on 22 August, but its artifact was not persisted in the active sandbox and was therefore not downloadable. This document is a reconstruction from the validated project state and the 22 August derivations, with only results that remain internally consistent retained.